RRAM-DP: Device-Calibrated Differential Privacy for In-Memory Edge Learning

Kwunhang Wong[1,2,*], Jichang Yang[1,2,*], Karl M.H. Lai[2], Hegan Chen[1,2], Songqi Wang[1,2], Wei Xuan[1], Ning Lin[2,4,†], Han Wang[2], Xiaojuan Qi[2,†] and Zhongrui Wang[1,3,†]

[1]ACCESS – AI Chip Center for Emerging Smart Systems, InnoHK Centers, Hong Kong Science Park, Hong Kong; [2]Department of Electrical and Computer Engineering, The University of Hong Kong, Hong Kong; [3]School of Microelectronics, Southern University of Science and Technology, Shenzhen, China; [4]Department of Electrical Engineering, City University of Hong Kong, Hong Kong

[*]Equal contribution to this work; [†]Corresponding authors: linning@hku.hk; xjqi@eee.hku.hk; wangzr@sustech.edu.cn

Abstract

Edge Artificial Intelligence of Things (AIoT) systems often collect sensitive data *in situ*, raising serious privacy concerns. Resistive-switching random-access memory (RRAM) is an attractive substrate for efficient AIoT thanks to its multi-bit storage and compute-in-memory (CiM) capabilities, while its inherently stochastic write behavior provides a natural source of randomness that can be leveraged for differential privacy (DP) protection. Yet how to transform this device-level randomness– typically viewed as detrimental to accuracy– into a principled randomized mechanism while preserving model utility remains underexplored. We propose RRAM-DP, a hardware–algorithm co-design that relaxes RRAM write–verify operations to inject calibrated noise for inherently (ε, δ)-DP with formal DP analysis; together with pretraining techniques, it renders a novel private, high-utility CiM-training paradigm. On CIFAR-10/100, STS-B, and SST-2, RRAM-DP-SGD incurs at best only a 3.8% accuracy drop at (ε=2,δ=O(1/n))-DP relative to non-private SGD. At the same privacy level, RRAM-DP-SGD delivers up to 57×& 3.2× energy savings and 2.7×& 1.8× speedups over A100 and DiVa-GEMM, respectively. These results point towards efficient, privacy-preserving in-memory training on RRAM at the edge.

CCS Concepts

• **Hardware → Memory and dense storage**; • **Security and privacy → Formal methods and theory of security**.

Keywords

RRAM, Memristor, AI accelerator, Compute-in-Memory, Differential Privacy, Edge AIoT, Device Variability

ACM Reference Format:

K. Wong, J. Yang, K. Lai, H. Chen, S. Wang, W. Xuan, N. Lin, H. Wang, X. Qi, and Z. Wang. 2026. RRAM-DP: Device-Calibrated Differential Privacy for In-Memory Edge Learning . In *IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26), November 08–12, 2026, San Jose, CA, USA.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3831252.3834125

1 Introduction

Edge Artificial Intelligence of Things (AIoT) systems increasingly operate *in situ* sensing and learning from local, privacy-sensitive environments to personalize services across modalities and tasks [45]. This shift heightens well-documented risks: seemingly anonymized records can be re-identified [31], and modern deep neural networks (DNNs) memorize and leak training data via membership inference attacks (MIAs) [7] and training data extraction attacks [6, 8].

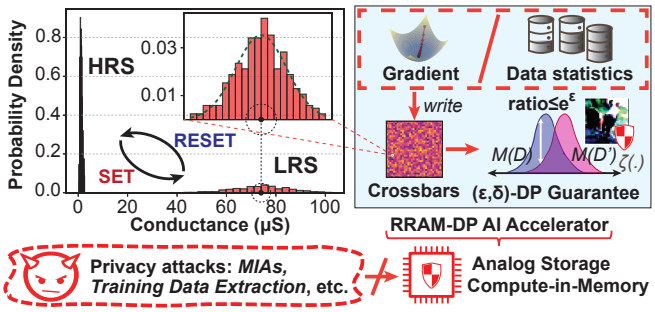


Figure 1: (LEFT) Histogram of cycle-to-cycle variation for an RRAM device between high resistance state (HRS) and low resistance state (LRS) with 300 iterative SET-RESET at fixed operation voltages. (RIGHT) Such noise can provide (ε, δ)-DP for storing sensitive data statistics/model parameters.

Differential privacy (DP) provides a principled defense by adding calibrated noise to query outputs, masking any single individual's contribution and thereby offering rigorous protection against database-level attacks [15]. DP-Stochastic Gradient Descent (SGD) extends DP guarantees to integrating noise injection in SGD optimizer, enabling privately trained DNNs with (ε, δ)-DP [1]. However, despite their widespread use, existing DP and DP-SGD frameworks still face critical challenges in AIoT applications as detailed below.

To date, most DP mechanisms are realized in *digital* hardware– namely CPUs/GPUs/TPUs and dedicated digital accelerators [32, 42]– which raises several tensions intersected at efficiency and privacy for edge AIoT: (i) pseudo-random noise sampling in digital platform has finite precision with rounding errors, which is susceptible to floating-point error attacks [21, 30]; (ii) repeated data movement between processors and memory for real-time randomized mechanism are energy-expensive in local DP [37] (iii) existing DP-SGD implementation libraries suffer $2\times - 1000\times$ in time and space complexity compared to normal SGD training [4], which are especially problematic in resource constrained edge.

Resistive-switching random-access memory (RRAM) is promising solution to address these challenges: First, RRAM's inherently stochastic programming offers a native source of entropy for true random number generation. Second, write operations of RRAM, originally regarded as a reliability problem, are naturally an in-place randomization with ***analog storage*** capability [39]. Thirdly, the ohmic nature of RRAM naturally allows efficient multiply-accumulates (MACs) computation, where RRAM-based Compute-in-memory (CiM) accelerator provides powerful ***in-memory vector-matrix multiplication***. Prior DP work explored memristor noise largely at a theoretical level, under the assumption of idealized i.i.d. Gaussian perturbations [18]. In practice, RRAM noise is non-identical, and practically difficult to be controlled across devices and cycles for

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ICCAD '26, San Jose, CA, USA

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ACM ISBN 979-8-4007-2873-0/2026/11
https://doi.org/10.1145/3831252.3834125